

Extremum-Normalized Stochastic Process Theory

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This paper develops Extremum-Normalized Stochastic Process Theory (ENSPT), a path-space framework in which a stochastic trajectory is represented relative to its realized extrema and their occurrence times. For a continuous process $S = (S_t)_{0 \leq t \leq 1}$, define

$$H = \max_{0 \leq t \leq 1} S_t, \quad L = \min_{0 \leq t \leq 1} S_t, \quad (1)$$

with extremum times τ_H and τ_L . The principal normalized processes are

$$h_t = \frac{S_t - t}{H - \tau_H}, \quad \ell_t = \frac{S_t - t}{L - \tau_L}. \quad (2)$$

Whenever the denominators are nonzero,

$$(H - \tau_H)h_t = (L - \tau_L)\ell_t = S_t - t, \quad (3)$$

so that

$$S_t = t + \frac{(H - \tau_H)h_t + (L - \tau_L)\ell_t}{2}. \quad (4)$$

The theory separates a realized stochastic trajectory into extremum geometry and normalized path shape. Brownian motion and geometric Brownian motion are used as principal examples. Killed transition kernels, boundary fluxes, bridge measures, Doob transforms, quadratic variation, functional data analysis, econometrics, finance, physics, biology, decision theory, philosophy, and computational learning are discussed as natural domains of interpretation or application. The emphasis is on exact pathwise identities, conditional probability laws, and statistically testable representations rather than on predictive claims.

The paper ends with “The End”

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1 Introduction

Financial and physical trajectories are commonly studied in their raw scale, yet many scientific questions concern shape rather than absolute magnitude. Normalization by an endogenous extremum is fundamentally different from normalization by an externally fixed scale: the normalizing quantity is itself a random functional of the complete trajectory.

The present theory formalizes this idea. Let

$$S = (S_t)_{0 \leq t \leq 1} \tag{5}$$

be a continuous stochastic process on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Define

$$H = \sup_{0 \leq u \leq 1} S_u, \quad L = \inf_{0 \leq u \leq 1} S_u, \quad (6)$$

and measurable extremum times

$$\tau_H \in \arg \max_{0 \leq u \leq 1} S_u, \quad \tau_L \in \arg \min_{0 \leq u \leq 1} S_u. \quad (7)$$

The basic transformed process is

$$X_t = S_t - t. \quad (8)$$

The high and low normalizations are then

$$h_t = \frac{X_t}{X_{\tau_H}}, \quad \ell_t = \frac{X_t}{X_{\tau_L}}. \quad (9)$$

The construction is motivated by stochastic-process theory [6, 8, 11], mathematical finance [3, 9, 12], and functional statistical analysis [10, 7].

2 Axiomatic Construction

Definition 2.1 (Extremum geometry). *The random vector*

$$\mathcal{G} = (H, L, \tau_H, \tau_L) \quad (10)$$

is called the extremum geometry of the path S .

Definition 2.2 (Extremum-normalized stochastic processes). *Provided*

$$(H - \tau_H)(L - \tau_L) \neq 0, \quad (11)$$

define

$$h_t = \frac{S_t - t}{H - \tau_H}, \quad \ell_t = \frac{S_t - t}{L - \tau_L}. \quad (12)$$

The processes h and ℓ are called the high-normalized and low-normalized processes, respectively.

Theorem 2.3 (Exact reconstruction). *For every realized path for which the normalizations are defined,*

$$(H - \tau_H)h_t = (L - \tau_L)\ell_t = S_t - t. \quad (13)$$

Consequently,

$$S_t = t + (H - \tau_H)h_t \quad (14)$$

and

$$S_t = t + (L - \tau_L)\ell_t. \quad (15)$$

Hence

$$S_t = t + \frac{(H - \tau_H)h_t + (L - \tau_L)\ell_t}{2}. \quad (16)$$

Proof. Substitution of the definitions gives

$$(H - \tau_H)h_t = (H - \tau_H) \frac{S_t - t}{H - \tau_H} = S_t - t. \quad (17)$$

The low-normalized identity follows identically. Averaging the two exact representations proves the final equation. $\square$

Corollary 2.4 (Exact proportionality). *Define*

$$R = \frac{L - \tau_L}{H - \tau_H}. \quad (18)$$

Then

$$h_t = R\ell_t. \quad (19)$$

Thus the pair (h, ℓ) has stochastic rank one: conditional on the realized extremum geometry, there is only one independent path factor.

3 Shape–Scale Decomposition

Define

$$A_H = H - \tau_H, \quad A_L = L - \tau_L. \quad (20)$$

Then

$$S_t = t + A_H h_t = t + A_L \ell_t. \quad (21)$$

This suggests the decomposition

$$\boxed{S. \longleftrightarrow (A_H, h.)} \quad (22)$$

or equivalently

$$\boxed{S. \longleftrightarrow (A_L, \ell.)} \quad (23)$$

The random scale and normalized path shape should not be confused with independent random variables. In general they are statistically dependent.

Proposition 3.1 (Scale invariance). *If X_t is replaced by cX_t for a nonzero constant c , then*

$$\frac{cX_t}{cX_{\tau_H}} = \frac{X_t}{X_{\tau_H}}. \quad (24)$$

Thus the normalized shape is invariant under multiplicative rescaling of the transformed process.

This property connects ENSPT to dimensional analysis and similarity methods in physics. The normalization removes one scale degree of freedom while retaining the path’s relative geometry.

4 Information and Filtrations

The variables H , L , τ_H , and τ_L generally depend on the entire sample path. Consequently, for $t < 1$,

$$H \notin \mathcal{F}_t \quad (25)$$

in general. Hence h_t and ℓ_t are not generally adapted to the original trading filtration.

Define the extremum sigma-field

$$\mathcal{E} = \sigma(H, L, \tau_H, \tau_L) \quad (26)$$

and the enlarged filtration

$$\mathcal{G}_t = \mathcal{F}_t \vee \mathcal{E}. \quad (27)$$

Then A_H and A_L are $\mathcal{G}_0$ -measurable, so h and ℓ are $\mathcal{G}_t$ -adapted.

This establishes an important distinction:

$$\boxed{\text{ex-post normalized path} \neq \text{real-time observable trading signal.}} \quad (28)$$

The distinction is central to econometric identification and to avoiding look-ahead bias in financial applications.

5 Diffusion Formulation

Let

$$dS_t = \mu_t dt + \sigma_t dW_t \quad (29)$$

be an Itô diffusion. Then

$$dX_t = (\mu_t - 1)dt + \sigma_t dW_t. \quad (30)$$

Conditional on the completed extremum geometry,

$$dh_t = \frac{\mu_t - 1}{A_H} dt + \frac{\sigma_t}{A_H} dW_t, \quad (31)$$

and

$$d\ell_t = \frac{\mu_t - 1}{A_L} dt + \frac{\sigma_t}{A_L} dW_t. \quad (32)$$

Their quadratic variations satisfy

$$[h]_t = \frac{[S]_t}{A_H^2}, \quad (33)$$

$$[\ell]_t = \frac{[S]_t}{A_L^2}, \quad (34)$$

$$[h, \ell]_t = \frac{[S]_t}{A_H A_L}. \quad (35)$$

Hence the instantaneous covariance matrix is

$$\Sigma_t = \sigma_t^2 \begin{pmatrix} A_H^{-2} & (A_H A_L)^{-1} \\ (A_H A_L)^{-1} & A_L^{-2} \end{pmatrix}. \quad (36)$$

Its determinant is zero:

$$\det(\Sigma_t) = 0. \quad (37)$$

Therefore the two normalized signals form a rank-one stochastic system.

6 Brownian Motion

Consider

$$S_t = s_0 + \mu t + \sigma W_t. \quad (38)$$

The transformed process is

$$X_t = S_t - t = s_0 + (\mu - 1)t + \sigma W_t. \quad (39)$$

The extremum geometry, however, is defined from S , not X :

$$H = \max_t S_t, \quad L = \min_t S_t. \quad (40)$$

This distinction is essential because, in general,

$$\arg \max_t S_t \neq \arg \max_t (S_t - t). \quad (41)$$

6.1 Killed transition kernel

For an interval (ℓ, h) , let $p_{\ell, h}(t; x, y)$ denote the transition density of the Brownian diffusion killed on leaving the interval. It has the spectral representation

$$p_{\ell, h}(t; x, y) = \frac{2}{h - \ell} e^{\frac{\mu(y-x)}{\sigma^2} - \frac{\mu^2 t}{2\sigma^2}} \sum_{n=1}^{\infty} e^{-\frac{n^2 \pi^2 \sigma^2 t}{2(h-\ell)^2}} \sin\left(\frac{n\pi(x-\ell)}{h-\ell}\right) \sin\left(\frac{n\pi(y-\ell)}{h-\ell}\right). \quad (42)$$

For zero drift this reduces to

$$p_{\ell, h}(t; x, y) = \frac{2}{h - \ell} \sum_{n=1}^{\infty} e^{-\frac{n^2 \pi^2 \sigma^2 t}{2(h-\ell)^2}} \sin\left(\frac{n\pi(x-\ell)}{h-\ell}\right) \sin\left(\frac{n\pi(y-\ell)}{h-\ell}\right). \quad (43)$$

These kernels are standard objects in diffusion theory and first-passage analysis [6, 4, 8].

6.2 Barrier-conditioned bridges

For $a < t < b$, conditional on

$$S_a = x, \quad S_b = y, \quad (44)$$

and survival inside (ℓ, h) , the bridge density is proportional to

$$p_{\ell, h}(t - a; x, z) p_{\ell, h}(b - t; z, y). \quad (45)$$

More precisely,

$$\mathbb{P}(S_t \in dz \mid S_a = x, S_b = y, \ell < S_u < h) = \frac{p_{\ell, h}(t - a; x, z) p_{\ell, h}(b - t; z, y)}{p_{\ell, h}(b - a; x, y)} dz. \quad (46)$$

This is the natural conditional path measure underlying the extremum-normalized Brownian construction.

7 Extremum Times and Boundary Flux

Ordinary killed kernels vanish at the absorbing boundaries:

$$p_{\ell,h}(t; x, \ell) = p_{\ell,h}(t; x, h) = 0. \quad (47)$$

First-passage distributions are consequently represented through probability fluxes. The probability current is

$$J(t, x) = \mu p(t, x) - \frac{\sigma^2}{2} \frac{\partial p(t, x)}{\partial x}. \quad (48)$$

At an absorbing boundary the density vanishes, leaving a normal derivative term. Thus boundary derivatives of the killed kernel encode the density of first contact.

For example, the lower and upper fluxes may be written schematically as

$$J_L(t; x) = \frac{\sigma^2}{2} \left. \frac{\partial p_{\ell,h}(t; x, y)}{\partial y} \right|_{y=\ell}, \quad (49)$$

$$J_H(t; x) = -\frac{\sigma^2}{2} \left. \frac{\partial p_{\ell,h}(t; x, y)}{\partial y} \right|_{y=h}. \quad (50)$$

The joint law of

$$(H, L, \tau_H, \tau_L) \quad (51)$$

is therefore most naturally represented through boundary-flux limits and excursion kernels rather than by treating the four variables as independent [4, 11].

7.1 Three-segment decomposition

If

$$\tau_L < \tau_H, \quad (52)$$

the path decomposes into

$$[0, \tau_L], \quad [\tau_L, \tau_H], \quad [\tau_H, 1]. \quad (53)$$

Its conceptual density structure is

$$\boxed{\text{initial-to-low flux} \times \text{low-to-high excursion} \times \text{high-to-terminal flux.}} \quad (54)$$

The reverse ordering gives

$$\boxed{\text{initial-to-high flux} \times \text{high-to-low excursion} \times \text{low-to-terminal flux.}} \quad (55)$$

This provides the appropriate mathematical representation of extremum geometry.

8 The Normalized Brownian Path

Define

$$A_H = H - \tau_H. \quad (56)$$

Then

$$Z_t = \frac{S_t - t}{A_H}. \quad (57)$$

At the high time,

$$Z_{\tau_H} = 1. \quad (58)$$

If $A_H > 0$, the constraints

$$L \leq S_t \leq H \quad (59)$$

become

$$\frac{L - t}{A_H} \leq Z_t \leq \frac{H - t}{A_H}. \quad (60)$$

Thus Z moves within a random translating corridor of constant width

$$\frac{H - L}{A_H}. \quad (61)$$

Introduce

$$Y_t = Z_t + \frac{t}{A_H}. \quad (62)$$

Then

$$Y_t = \frac{S_t}{A_H} \quad (63)$$

and the corridor becomes stationary:

$$\frac{L}{A_H} \leq Y_t \leq \frac{H}{A_H}. \quad (64)$$

9 Geometric Brownian Motion

Consider

$$dS_t = \mu S_t dt + \sigma S_t dW_t. \quad (65)$$

By Itô's lemma,

$$Y_t = \log S_t \quad (66)$$

satisfies

$$dY_t = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW_t. \quad (67)$$

Therefore GBM becomes Brownian motion with drift in logarithmic coordinates [3, 9, 12].

Let

$$M = \max_t Y_t, \quad m = \min_t Y_t. \quad (68)$$

Then

$$H = e^M, \quad L = e^m. \quad (69)$$

Because the logarithm is monotone,

$$\tau_H = \arg \max_t Y_t, \quad \tau_L = \arg \min_t Y_t. \quad (70)$$

Consequently, the extremum geometry of GBM is obtained from the extremum geometry of drifted Brownian motion through

$$(L, H) = (e^m, e^M). \quad (71)$$

The corresponding density transformation is

$$f_{L,H,\tau_L,\tau_H}(\ell, h, \lambda, \tau) = \frac{1}{\ell h} f_{m,M,\tau_L,\tau_H}(\log \ell, \log h, \lambda, \tau). \quad (72)$$

The original ENSPT process remains

$$Z_t = \frac{e^{Y_t} - t}{e^M - \tau_H}. \quad (73)$$

Thus the GBM problem consists of a linear diffusion problem in log-space followed by a nonlinear push-forward to price space.

10 Quadratic Variation and Volatility

For GBM,

$$Z_t = \frac{S_t - t}{H - \tau_H}. \quad (74)$$

Since

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad (75)$$

we obtain

$$dZ_t = \frac{\mu S_t - 1}{H - \tau_H} dt + \frac{\sigma S_t}{H - \tau_H} dW_t. \quad (76)$$

Consequently,

$$[Z]_t = \frac{\sigma^2}{(H - \tau_H)^2} \int_0^t S_u^2 du. \quad (77)$$

For the Brownian model,

$$[Z]_t = \frac{\sigma^2 t}{(H - \tau_H)^2}. \quad (78)$$

Thus extremum normalization changes volatility scale but preserves the underlying temporal organization of the stochastic variation.

This connects ENSPT naturally with realized-volatility econometrics [1, 2].

11 Probability Law on Path Space

Let

$$\mathcal{G} = (H, L, \tau_H, \tau_L). \quad (79)$$

The conditional normalized-path law is

$$\mathbb{P}_Z(dz \mid \mathcal{G} = g). \quad (80)$$

The unconditional law is the mixture

$$\boxed{\mathbb{P}_Z(dz) = \int \mathbb{P}_Z(dz \mid \mathcal{G} = g) \mathbb{P}_{\mathcal{G}}(dg).} \quad (81)$$

For any measurable functional

$$F : C[0, 1] \rightarrow \mathbb{R}, \quad (82)$$

$$\mathbb{E}[F(Z)] = \int \mathbb{E}[F(Z) \mid \mathcal{G} = g] \mathbb{P}_{\mathcal{G}}(dg). \quad (83)$$

This is the central probabilistic statement of ENSPT.

The theory therefore separates two kinds of uncertainty:

$$\boxed{\text{extremum-geometry uncertainty} + \text{conditional path-shape uncertainty}.} \quad (84)$$

The law of total variance gives

$$\text{Var}(F(Z)) = \mathbb{E}[\text{Var}(F(Z) \mid \mathcal{G})] + \text{Var}[\mathbb{E}(F(Z) \mid \mathcal{G})]. \quad (85)$$

Similarly,

$$\text{Cov}(Z_s, Z_t) = \mathbb{E}[\text{Cov}(Z_s, Z_t \mid \mathcal{G})] \quad (86)$$

$$+ \text{Cov}(\mathbb{E}[Z_s \mid \mathcal{G}], \mathbb{E}[Z_t \mid \mathcal{G}]). \quad (87)$$

12 The Extremum Ratio

Define

$$R = \frac{L - \tau_L}{H - \tau_H}. \quad (88)$$

Then

$$h_t = R\ell_t. \quad (89)$$

The ratio R is a random functional of the entire trading path.

If

$$A_H = H - \tau_H, \quad A_L = L - \tau_L, \quad (90)$$

then

$$R = \frac{A_L}{A_H}. \quad (91)$$

The transformation

$$(A_H, A_L) \longrightarrow (A_H, R) \quad (92)$$

has Jacobian

$$\left| \frac{\partial(A_H, A_L)}{\partial(A_H, R)} \right| = |A_H|. \quad (93)$$

Hence the density of R is formally

$$f_R(r) = \int \int \int |a| f_{A_H, A_L, \tau_H, \tau_L}(a, ra, \tau, \lambda) da d\tau d\lambda. \quad (94)$$

This random variable summarizes the proportional relationship between the high-normalized and low-normalized processes.

13 Economics and Finance

In financial economics, the framework provides a distinction between *level*, *scale*, *extremum geometry*, and *intraday shape*. This is complementary to standard continuous-time asset-pricing models [9, 12].

For an observed trading day d ,

$$Z_{d,t} = \frac{S_{d,t} - t}{H_d - \tau_{H,d}}. \quad (95)$$

Across N trading days,

$$\hat{m}_Z(t) = \frac{1}{N} \sum_{d=1}^N Z_{d,t} \quad (96)$$

estimates the population mean normalized trajectory.

A functional covariance estimator is

$$\hat{K}_Z(s, t) = \frac{1}{N-1} \sum_{d=1}^N (Z_{d,s} - \hat{m}_Z(s)) (Z_{d,t} - \hat{m}_Z(t)). \quad (97)$$

This permits functional principal-component analysis [10].

Potential empirical questions include:

1. whether normalized intraday shapes are stable across market regimes;
2. whether high-first and low-first days have different shape distributions;
3. whether extremum geometry explains a significant fraction of normalized path variance;
4. whether Brownian and GBM models generate empirically distinguishable normalized path laws;
5. whether volatility clustering survives extremum normalization.

None of these questions should be answered by imposing the model a priori; they are empirical hypotheses.

14 Econometrics

ENSPT naturally generates a functional-data panel:

$$\{Z_{d,t} : d = 1, \dots, N, t \in [0, 1]\}. \quad (98)$$

A generic functional regression may be written as

$$Y_d = \alpha + \int_0^1 \beta(t) Z_{d,t} dt + \gamma^\top X_d + \varepsilon_d. \quad (99)$$

The extremum geometry may enter as a separate regressor vector:

$$X_d^{\text{ext}} = (H_d, L_d, \tau_{H,d}, \tau_{L,d}, R_d). \quad (100)$$

One can then distinguish

$$\text{shape effects} \quad (101)$$

from

$$\text{extremum-geometry effects.} \quad (102)$$

Care is required because the normalization is ex post. Any forecasting exercise must avoid using future extrema when constructing a predictor at time t .

15 Statistics and Functional Data Analysis

The normalized trajectory is a random element of

$$C[0, 1]. \quad (103)$$

Let

$$m_Z(t) = \mathbb{E}[Z_t] \quad (104)$$

and

$$K_Z(s, t) = \mathbb{E}[(Z_s - m_Z(s))(Z_t - m_Z(t))]. \quad (105)$$

The associated covariance operator is

$$(\mathcal{K}f)(t) = \int_0^1 K_Z(s, t) f(s) ds. \quad (106)$$

Under standard compactness conditions, its eigenfunctions satisfy

$$\mathcal{K}\phi_k = \lambda_k \phi_k. \quad (107)$$

The Karhunen–Loève representation is

$$Z_t = m_Z(t) + \sum_{k=1}^{\infty} \xi_k \phi_k(t), \quad (108)$$

with

$$\mathbb{E}[\xi_k] = 0, \quad \mathbb{E}[\xi_j \xi_k] = \lambda_k \mathbf{1}_{\{j=k\}}. \quad (109)$$

This gives a finite-dimensional approximation when the leading eigenvalues dominate.

16 Physics: Similarity and Scaling

The decomposition

$$S_t = t + A_H Z_t \quad (110)$$

resembles a similarity transformation: a scale variable multiplies a dimensionless shape variable.

This is conceptually related to nondimensionalization, self-similarity, and renormalization ideas in mathematical physics [6]. The analogy should not be interpreted as asserting that financial markets literally obey a physical renormalization group.

The dimensionless range

$$\frac{H - L}{\sigma} \quad (111)$$

and the extremum ratio

$$R = \frac{L - \tau_L}{H - \tau_H} \quad (112)$$

are natural scale-free quantities.

17 Biology and Other Dynamical Systems

The mathematical construction is not inherently financial. Suppose X_t denotes population size, biomass, temperature, concentration, or another continuous dynamical quantity. Define

$$M = \max_t X_t, \quad m = \min_t X_t. \quad (113)$$

An extremum-normalized trajectory can be defined by

$$Z_t = \frac{X_t - c(t)}{M - c(\tau_M)}, \quad (114)$$

where $c(t)$ is a deterministic reference trajectory.

The same shape-scale decomposition then applies. This makes ENSPT potentially relevant to biological growth curves, physiological signals, ecological trajectories, and laboratory time series, subject to domain-specific measurement assumptions.

18 Psychology and Psychiatry

In psychology and psychiatry, trajectories of observed quantities such as behavioral scores, reaction times, or symptom-rating scales may be modeled as stochastic processes. An extremum-normalized representation could be used to compare trajectory shapes after removing one endogenous scale.

However, normalization must not be interpreted as a diagnostic operation. Clinical inference requires validated measurement models, appropriate longitudinal designs, and domain-specific causal assumptions.

Thus ENSPT supplies a mathematical representation, not a psychiatric theory.

19 Philosophy and Epistemology

The construction has a useful epistemological interpretation.

The completed path contains two distinct kinds of information:

$$\text{what the trajectory did} \quad (115)$$

and

$$\text{where its extrema occurred.} \quad (116)$$

Conditioning on the extrema changes the information set. Hence

$$\mathbb{P}(Z_t \mid \mathcal{G}) \quad (117)$$

is an explicitly information-dependent object.

This provides a mathematical illustration of the distinction between *ex ante* and *ex post* knowledge: a completed trajectory can be normalized by information that was not available at intermediate times.

The construction therefore provides a precise language for discussing look-ahead, hindsight, conditioning, and representation without confusing them with prediction.

20 Religion and Interpretive Thought

The framework may also be used metaphorically in philosophical or religious reflection. A trajectory can be viewed as possessing both a realized path and a geometry of extrema; normalization then asks how the path looks relative to its own boundaries.

Such interpretations belong to hermeneutics rather than stochastic-process theory. The mathematical result remains the pathwise identity

$$S_t = t + (H - \tau_H)Z_t. \quad (118)$$

No theological proposition follows mathematically from ENSPT.

21 Political Science, Geopolitics, and International Relations

The same mathematical vocabulary can describe abstract capability or resource trajectories:

$$C_t \quad (119)$$

may represent a suitably measured institutional, economic, or technological quantity, while

$$C_{\max}, \quad C_{\min}, \quad \tau_{\max}, \quad \tau_{\min} \quad (120)$$

describe its realized extrema.

An extremum-normalized representation can then compare trajectory shape across different units whose absolute scales differ.

This is a descriptive methodology, not a theory of political behavior. Political interpretation requires institutional, historical, legal, and causal evidence in addition to the stochastic representation.

22 Welfare Economics

Let W_t denote an aggregate welfare index. Its normalized trajectory can be studied through

$$Z_t^W = \frac{W_t - c(t)}{W_{\max} - c(\tau_{\max})}. \quad (121)$$

The decomposition separates:

$$\text{welfare scale} \quad (122)$$

from

$$\text{normalized welfare trajectory}. \quad (123)$$

This does not resolve interpersonal comparability or welfare measurement problems; those remain substantive issues in welfare economics.

23 Warfare Economics and Strategic Resources

For an abstract resource process Q_t , such as a suitably defined production or logistical capacity index, the same mathematics yields

$$Q_t = c(t) + (Q_{\max} - c(\tau_{\max}))Z_t^Q. \quad (124)$$

The framework can therefore distinguish the scale of a resource system from the normalized temporal pattern of its realized fluctuations.

This is deliberately a high-level analytical representation. It does not provide operational military instructions or targeting procedures.

24 Computer Science and Algorithms

The theory can be implemented by the following abstract procedure.

INPUT:

time grid $t[0], \dots, t[n]$
observed trajectory $S[0], \dots, S[n]$

1. Compute $H = \max(S)$
2. Compute $L = \min(S)$
3. Find $\tau_H = \text{time of } H$
4. Find $\tau_L = \text{time of } L$
5. Compute $A_H = H - \tau_H$
6. Compute $A_L = L - \tau_L$
7. For each time t :
 $h[t] = (S[t] - \tau_H) / A_H$
 $l[t] = (S[t] - \tau_L) / A_L$
8. Verify:
 $\text{reconstruction}[t] =$
 $t + (A_H * h[t] + A_L * l[t]) / 2$
9. Compute residual:
 $D[t] = \text{reconstruction}[t] - S[t]$
10. Return:
 $H, L, \tau_H, \tau_L,$
 h, l, D

In exact arithmetic,

$$D_t = 0. \quad (125)$$

With observed data, nonzero values of D_t diagnose numerical error, timestamp mismatch, interpolation, or inconsistent extrema definitions.

25 Data Science and Machine Learning

The normalized path can be represented by a feature vector

$$\mathbf{x} = (\xi_1, \dots, \xi_K, H, L, \tau_H, \tau_L, R), \quad (126)$$

where ξ_k are functional principal-component scores.

This provides a dimensionality-reduction pipeline:

$$\begin{array}{ccccccc} \text{raw} & \rightarrow & \text{extremum} & \rightarrow & \text{normalized} & \rightarrow & \text{functional} \\ \text{trajectory} & & \text{geometry} & & \text{path} & & \text{basis} \\ & & & & & & \text{low-dimensional} \\ & & & & & & \text{representation} \end{array} \quad (127)$$

A generic machine-learning model may then take the form

$$\hat{Y} = f_{\theta}(H, L, \tau_H, \tau_L, \xi_1, \dots, \xi_K). \quad (128)$$

The representation can be used with regression, clustering, anomaly detection, Gaussian processes, recurrent models, state-space models, or transformer-based sequence models. The normalization should remain separate from the predictive model so that leakage can be controlled.

26 Pseudo-Code for a Model-Based Simulation

A model-based ENSPT simulation can be expressed as follows.

```

INPUT:
  stochastic model for S
  number of simulated paths N
  time grid  $t[0], \dots, t[n]$ 

FOR d = 1, \dots, N:

  Simulate  $S_d$  on  $[0,1]$ 

   $H_d$       = maximum of  $S_d$ 
   $L_d$       = minimum of  $S_d$ 
   $\tau_{H_d}$  = time of  $H_d$ 
   $\tau_{L_d}$  = time of  $L_d$ 

   $A_{H_d} = H_d - \tau_{H_d}$ 
   $A_{L_d} = L_d - \tau_{L_d}$ 

  IF  $A_{H_d}$  and  $A_{L_d}$  are nonzero:

    FOR each t:
       $Z_d[t] = (S_d[t] - t) / A_{H_d}$ 
       $Lsig_d[t] = (S_d[t] - t) / A_{L_d}$ 

     $R_d = A_{L_d} / A_{H_d}$ 

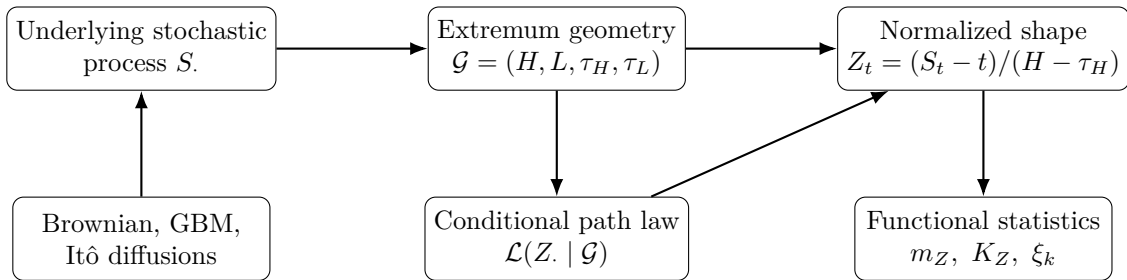
    Verify:
       $Z_d[t] = R_d * Lsig_d[t]$ 

END FOR

Estimate:
  mean normalized path
  covariance function
  distribution of R
  distribution of extremum geometry
  functional principal components
  model-vs-data discrepancy

```

27 A Unified Conceptual Diagram



The diagram emphasizes the hierarchical nature of ENSPT:

$$\boxed{\text{diffusion} \rightarrow \text{extremum geometry} \rightarrow \text{conditional shape} \rightarrow \text{statistical representation.}} \quad (129)$$

28 Interdisciplinary Synthesis

Field	ENSPT object	Interpretation
Stochastic calculus	$S_t, H, L, \tau_H, \tau_L$	Path-dependent random variables and diffusion functionals.
Mathematical finance	Z_t	Scale-normalized intraday price trajectory.
Econometrics	$Z_{d,t}$	Functional panel observation with endogenous ex-post normalization.
Statistics	$m_Z(t), K_Z(s, t)$	Population mean and covariance of normalized paths.
Physics	$S_t = t + A_H Z_t$	Similarity and scale–shape decomposition.
Biology	$X_t - c(t)$	Normalized trajectory of a biological dynamical system.
Computer science	$(H, L, \tau_H, \tau_L, Z)$	Structured representation of a time series.
AI/ML	$(\xi_1, \dots, \xi_K, \mathcal{G})$	Low-dimensional representation for learning.
Philosophy	$\mathcal{L}(Z \mid \mathcal{G})$	Formal distinction between realized and information-conditioned descriptions.
Welfare economics	Z_t^W	Shape-normalized welfare trajectory.
Political science	Z_t^C	Descriptive normalization of institutional or capability trajectories.
International relations	$\mathcal{G}$	Geometry of observed aggregate trajectories, not a causal theory.
Warfare economics	Z_t^Q	Abstract normalization of resource-capacity trajectories.
Psychology	Z_t	Possible longitudinal shape representation, subject to measurement validity.
Religion	Shape–scale analogy	Interpretive metaphor rather than a mathematical theological claim.

29 Model Comparison

Two benchmark models are particularly useful.

29.1 Brownian model

$$dS_t = \mu dt + \sigma dW_t. \quad (130)$$

29.2 Geometric Brownian model

$$dS_t = \mu S_t dt + \sigma S_t dW_t. \quad (131)$$

Their normalized laws are

$$\mathbb{P}_Z^B \quad (132)$$

and

$$\mathbb{P}_Z^{GBM}. \quad (133)$$

An empirical dataset generates

$$\widehat{\mathbb{P}}_Z. \quad (134)$$

The model comparison problem is therefore

$$\widehat{\mathbb{P}}_Z \text{ versus } \mathbb{P}_Z^B, \quad \widehat{\mathbb{P}}_Z \text{ versus } \mathbb{P}_Z^{GBM}. \quad (135)$$

Suitable discrepancies include integrated squared error, Wasserstein distance on discretized paths, energy distance, kernel two-sample statistics, or likelihood-based procedures when tractable.

30 Empirical Identification

A practical empirical study should distinguish four stages:

1. **Path construction:** obtain synchronized intraday observations and normalize time to $[0, 1]$.
2. **Extremum extraction:** compute H, L, τ_H, τ_L using a prespecified convention.
3. **Shape construction:** compute h_t, ℓ_t, Z_t without using any future information if the object is intended for real-time forecasting.
4. **Distributional analysis:** estimate the empirical law of extremum geometry and normalized shape.

The distinction between ex-post descriptive analysis and ex-ante forecasting must be maintained throughout.

31 Limitations

Several mathematical and empirical limitations deserve explicit treatment.

First, the normalization fails when

$$H - \tau_H = 0 \tag{136}$$

or

$$L - \tau_L = 0. \tag{137}$$

Second, extremum times can be nonunique for processes with flat segments or discrete observations. A deterministic tie-breaking rule is then required.

Third, the observed high and low in discrete financial data are not necessarily the continuous-path extrema.

Fourth, microstructure noise can materially affect high-frequency extrema.

Fifth, ex-post normalization creates look-ahead information and must not be used naively in predictive trading models.

Sixth, Brownian motion and GBM are benchmark models rather than empirical truths. Real financial data exhibit features such as stochastic volatility, jumps, leverage effects, volatility clustering, market microstructure effects, and non-Gaussian returns [5, 1, 2].

32 The Central Theorem of ENSPT

The entire construction can be summarized in the following statement.

Theorem 32.1 (Extremum-normalized stochastic decomposition). *Let $S = (S_t)_{0 \leq t \leq 1}$ be a continuous stochastic process. Suppose that H, L, τ_H , and τ_L are well-defined and*

$$(H - \tau_H)(L - \tau_L) \neq 0. \tag{138}$$

Define

$$Z_t = \frac{S_t - t}{H - \tau_H} \tag{139}$$

and

$$R = \frac{L - \tau_L}{H - \tau_H}. \tag{140}$$

Then

$$S_t = t + (H - \tau_H)Z_t, \tag{141}$$

$$\ell_t = \frac{Z_t}{R}, \tag{142}$$

$$h_t = Z_t. \tag{143}$$

Consequently, the realized stochastic trajectory is represented by the pair

$$(H, L, \tau_H, \tau_L; Z.), \quad (144)$$

and its probability law admits the mixture representation

$$\mathbb{P}_Z(dz) = \int \mathbb{P}_Z(dz \mid \mathcal{G} = g) \mathbb{P}_{\mathcal{G}}(dg). \quad (145)$$

Proof. The reconstruction identity follows immediately from the definition of Z_t . The relation for ℓ_t follows from

$$\ell_t = \frac{S_t - t}{L - \tau_L} = \frac{H - \tau_H}{L - \tau_L} Z_t = \frac{Z_t}{R}. \quad (146)$$

The mixture representation follows from the law of total probability on the path space. $\square$

33 Research Program

The theory suggests the following research program:

1. derive numerical representations of the joint extremum law for selected diffusions;
2. derive high-accuracy simulation methods for extremum-conditioned paths;
3. estimate the empirical law of $\mathcal{G}$ from high-frequency financial data;
4. estimate the covariance operator of normalized paths;
5. perform functional principal-component analysis;
6. compare Brownian, GBM, stochastic-volatility, and jump-diffusion models;
7. investigate whether normalized shape distributions are stable across assets and market regimes;
8. develop leakage-free online approximations for applications requiring real-time information;
9. study extensions to multidimensional stochastic processes;
10. investigate noncontinuous processes, including jump processes, with appropriate definitions of extrema and hitting times.

A particularly important extension is stochastic volatility:

$$dS_t = \mu_t S_t dt + \sigma_t S_t dW_t, \quad (147)$$

$$d\sigma_t = a(\sigma_t, t)dt + b(\sigma_t, t)dB_t. \quad (148)$$

Another is a jump-diffusion model:

$$dS_t = \mu_t dt + \sigma_t dW_t + \int_{\mathbb{R}} x \tilde{N}(dt, dx). \quad (149)$$

In these settings, the extrema may occur through jumps, and the continuous boundary-flux theory must be replaced or supplemented by jump-crossing probabilities.

34 Conclusion

Extremum-Normalized Stochastic Process Theory begins with the elementary identities

$$h_t = \frac{S_t - t}{H - \tau_H}, \quad \ell_t = \frac{S_t - t}{L - \tau_L}, \quad (150)$$

but the consequences are substantially broader.

The fundamental reconstruction is

$$\boxed{S_t = t + (H - \tau_H)h_t = t + (L - \tau_L)\ell_t.} \quad (151)$$

The pair of signals is therefore not a two-factor stochastic system. Rather, it is two normalizations of the same underlying path factor:

$$\boxed{h_t = R\ell_t.} \quad (152)$$

The complete stochastic structure is represented by

$$\boxed{\mathfrak{Z} = (H, L, \tau_H, \tau_L, Z.), \quad Z_t = \frac{S_t - t}{H - \tau_H}.} \quad (153)$$

Its law separates naturally into

$$\boxed{\text{extremum geometry} + \text{conditional normalized path shape.}} \quad (154)$$

For Brownian motion, this leads to killed heat kernels, boundary fluxes, excursion measures, bridges, and Doob transforms. For geometric Brownian motion, logarithmic transformation reduces the extremum geometry to that of drifted Brownian motion, after which the nonlinear price transformation recovers the normalized financial path.

The broader significance of ENSPT is therefore not that it supplies a new price predictor. Its primary contribution is representational: it constructs a probability theory for the *shape of a completed stochastic trajectory relative to its own realized extremal geometry*. That object can then be studied with stochastic calculus, statistics, econometrics, functional data analysis, machine learning, and domain-specific dynamical models.

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35 Glossary

Brownian motion

A continuous Gaussian stochastic process with independent stationary increments.

Doob h -transform

A transformation of a Markov process used to construct conditioned processes from a positive harmonic or excessive function.

Extremum geometry

The random vector

$$(H, L, \tau_H, \tau_L). \quad (155)$$

Extremum-normalized process

A stochastic trajectory normalized by its realized extremum and extremum time.

Functional data

Observations whose statistical units are curves or functions rather than finite-dimensional vectors.

Killed diffusion

A diffusion stopped or terminated upon reaching a prescribed boundary.

Quadratic variation

The limiting accumulated squared increments of a stochastic process.

Shape-scale decomposition

The representation

$$S_t = t + A_H Z_t \quad (156)$$

in which A_H is a random scale and Z_t is a normalized shape.

Stochastic bridge

A stochastic process conditioned on values at one or more endpoints.

Stochastic process

A family of random variables indexed by time or another parameter.

Path space

The space of possible sample trajectories, here typically $C[0, 1]$.

Boundary flux

The probability current through an absorbing boundary, used to characterize first-passage events.

Look-ahead bias

Bias arising when information unavailable at a decision time is used in a forecast or estimation procedure.

Geometric Brownian motion

The diffusion

$$dS_t = \mu S_t dt + \sigma S_t dW_t. \quad (157)$$

Extremum ratio

The random variable

$$R = \frac{L - \tau_L}{H - \tau_H}. \quad (158)$$

Karhunen–Loève expansion

The orthogonal eigenfunction representation of a second-order stochastic process based on its covariance operator.

The End